Evaluation and Comparison of Machine Learning Models on Fine Needle Aspiration Data for Breast Cancer Diagnosis

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I. Introduction

Breast cancer remains a leading cause of cancer-related mortality among women globally, accounting for 15% of all cancer deaths in women [1]. Breast cancer is the most commonly diagnosed cancer among women, causing an estimated 43,700 deaths in the United States in 2023. These statistics underscore the significant health burden posed by breast cancer, affecting a wide age range of women and leading to substantial morbidity and mortality. Early detection is crucial for improving outcomes, as patients with localized breast cancer have a 99% chance of surviving for at least five years, compared to only a 30% survival rate for those with metastatic cancer [2]. Early diagnosis facilitates timely treatment and significantly improves prognosis, underscoring the need for effective diagnostic strategies.

Currently, the most common diagnostic method is digital mammography (DM), a two-dimensional imaging technique that uses low-dose X-rays to identify abnormal masses in breast tissue. While digital mammography has advanced beyond traditional physical examination, it is not without its challenges. Manual review of mammograms by radiologists is susceptible to human error [3]. To address these challenges, there has been a growing increase in the use of machine-assisted imaging and machine-learning approaches to improve diagnostic accuracy. One such advancement is Digital Breast Tomosynthesis (DBT), which uses X-rays from multiple angles to create three-dimensional reconstructions of breast tissue. This approach has been shown to enhance cancer detection, especially in women with dense breast tissue [4]. However, DBT involves higher doses of X-ray exposure, raising concerns about potential DNA damage and increased cancer risk. Additionally, its higher cost and limited accessibility restrict widespread use. These limitations highlight the need to explore alternative diagnostic methods.

Fine-needle aspiration (FNA) offers a minimally invasive and cost-effective alternative to imaging-based methods. FNA involves using a thin, hollow needle to extract cells from a suspicious breast lump, which is then sent to a pathologist for examination. The extracted cells are assessed for various features, including cell size, shape, and distribution, to help pathologists differentiate between benign and malignant conditions. Recently, there has been increasing interest in integrating computer-assisted techniques with FNA to improve diagnostic accuracy. While still mainly used in research, methods like Support Vector Machines (SVM) have shown significant potential, achieving up to 92.7% accuracy in distinguishing between benign and malignant cells [5].

Although previous studies have primarily focused on simpler classifiers and ensemble methods like SVM and Random Forests (RF), there has been limited exploration of more complex models, particularly Neural Networks (NN), in FNA. NNs have shown outstanding performance in complex image analysis tasks, such as accurately diagnosing neurological conditions from three-dimensional cerebral angiographic images [6]. Consequently, previous research has often overlooked the potential benefits of integrating more advanced models to capture intricate patterns in cytological data.

To address these gaps, our study investigates a range of different models, from simpler algorithms like K-Nearest Neighbors (KNN) to advanced Neural Networks. By evaluating and optimizing the performance of each model, we aim to identify the most effective machine-learning approach that balances complexity and accuracy. Specifically, we employ KNN as the simplest model, RF as a medium-complexity ensemble method, and NNs as the most complex model. Ultimately, our goal is to find the best machine-learning model for distinguishing between benign and malignant breast conditions, contributing to the development of more reliable and efficient diagnostic tools. By expanding the range of models and optimizing their configurations, we aim to contribute to further advancements in computer-assisted FNA diagnostics.

II. Methods

A. Data

Our dataset was taken from a 1993 paper on nuclear feature extraction on digitized FNA images [7]. This dataset contains features extracted from FNA images from breast masses. It contains 569 records, each with 30 features extracted from digitized FNA images of breast masses. The features describe characteristics of the cell nuclei, including attributes like radius, texture, and area, with values for mean and standard error. Each record is labeled as “benign” or “malignant.” However, as with any dataset, it may still contain some noise or missing data, which we address during pre-processing.

Before the analysis, we performed several pre-processing steps on the dataset. First, the target variable (“diagnosis”) was encoded into binary values: “M” for malignant was transformed to 1, and “B” for benign to 0. The “id” column was dropped as it was irrelevant to classification. We then standardized the features using StandardScaler to ensure

equal weight for each feature, as some had larger numerical ranges, which could otherwise skew the model's predictions.

To enhance model performance and avoid overfitting, we applied Recursive Feature Elimination (RFE) with a Random Forest Classifier as the base estimator. RFE is a feature selection technique that recursively removes the least important features, allowing us to focus on the most relevant ones. This method is particularly effective in high-dimensional datasets. Previous studies have shown that RFE, when applied to FNA data, effectively improves prediction accuracy and reduces computational load [8]. Based on the RFE output, we selected the top 10 features which will be used in our experiment. Our selected features are displayed in Table I.

Feature	Description	Mean	Standard Deviation
1) Perimeter mean	Perimeter of the cell shape	122.80 mm	24.29 mm
2) Area mean	Area of the cell	1001.0 mm^2	354.00 mm^2
3) Concavity mean	Severity of concave portions of the contour	0.3001	0.14710
4) Concave points mean	Number of concave portions of the contour	0.14710	0.07017
5) Radius worst	Worst case radius value	25.38 mm	6.47 mm
6) Texture worst	Worst case texture value	17.33	4.18
7) Perimeter worst	Worst case perimeter value	184.60 mm	45.07 mm
8) Area worst	Worst case area value	2019.0 mm^2	617.00 mm^2
9) Concavity worst	Worst case concavity value	0.7119	0.1680
10) Concave points worst	Worst case concave points value	0.2654	0.0870

TABLE I: List of the selected features used in the experiment and their measured values.

B. Overview

In this study, we aim to build and compare the performance of our selected machine learning models: KNN, RF, and NN. Each model will be trained to classify breast tumors as either benign or malignant using FNA data.

We will begin by optimizing and adjusting the structure of each model, taking into account our feature selection technique, which may lead to a different set of features than previous studies. After fine-tuning the models, we will compare their performance using the most optimal settings. Each model will output a binary classification of benign or malignant, and performance will be evaluated using metrics such as accuracy, precision, recall, and F1-score.

C. Models

*1) **K-Nearest Neighbor (KNN):*** KNN is a simple, distance-based classifier that assigns a label based on the majority vote of its k nearest neighbors. It does not make assumptions about the underlying data distribution and works by directly comparing the input sample to the entire training dataset. Chosen for its simplicity, KNN is effective as a baseline model, especially for smaller datasets with distinct feature patterns, providing a good starting point before more complex models are used.

Training and Prediction: KNN does not require a formal training phase, as it simply memorizes the training data. During the prediction phase, the algorithm computes the Euclidean distance between the test point and every other point in the training dataset. The algorithm selects the k nearest neighbors based on the calculated distance and assigns the most common class label as the prediction.

Euclidean distance:

$$d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

where x_i and y_i are the feature values of the test and training samples, respectively, i represents a specific feature, and n is the total number of features.

Parameters: The key parameter for KNN is the number of neighbors, k, which determines how many neighbors to consider for prediction. Smaller values of k can make the model sensitive to noise, while larger values may smooth local variations, leading to overfitting.

*2) **Random Forest (RF):*** A Random Forest is an ensemble model that combines several decision trees (weak learners) to make a final prediction. This technique is chosen for its robustness to noise and ability to capture complex feature interactions, such as texture and thickness, critical for distinguishing benign from malignant tumors. By aggregating predictions from multiple trees, RF reduces overfitting and improves accuracy, making it well-suited for the high-dimensional FNA dataset.

Training and Prediction: During training, The Random Forest model is trained by building multiple decision trees that are independently trained on random subsets of the data. For the prediction phase, each tree votes on the sample's label, and the final result outputs the majority vote (mode) across all trees, classifying as either benign and malignant tumors.

Parameters: For RF trees we will focus on 2 parameters, n-estimators (number of trees) and maximum depth, which limits the depth of each tree.

*3) **Neural Networks (NNs):*** Neural Networks consist of an input layer, multiple hidden layers, and an output layer. Each neuron processes input data through weighted connections, which are adjusted during training to learn complex, non-linear patterns in the data. NNs are highly flexible and capable of modeling complex relationships in data, making them a potentially suitable for tasks like FNA classification, capable of capturing more intricate patterns may exist between more complex features.

Training and Prediction: During training, input data is passed through the network in a forward pass, where activation functions (like ReLU) capture non-linear patterns. Back-propagation then adjusts the weights using gradient

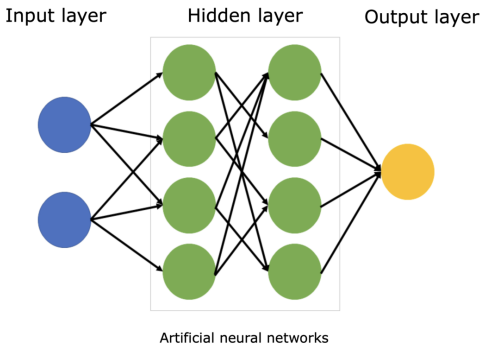


Fig. 1: Diagram of an artificial neural network showing the input, hidden, and output layers.

descent to minimize the error. For prediction, the trained model makes predictions by performing a forward pass on new test data.

Parameters: Neural networks can use an arbitrary number of hidden neurons and hidden layers. We will optimize our model for the most accurate number of neurons and layers.

D. Experimental Design

To comprehensively evaluate the performance of each model (KNN, RF, NN) in classifying FNA-based breast cancer samples as benign or malignant, we designed two experiments:

*1) **Model Optimization and Generalization Testing**:* The first experiment focuses on optimizing each model's parameters to achieve the highest classification accuracy while assessing their generalizability. A manual parameter search is conducted by iterating over predefined ranges of parameter values for each model, chosen based on their frequent use in previous studies and standard practices in machine learning literature.

Specifically, the following key parameters are tuned:

- **KNN**: Number of neighbors considered during classification (**k**), varied from 1 to 50.
- **RF**: The number of trees is tested between 1 and 250, and the maximum tree depth is tested between 1 and 30
- **NN**: The number of neurons per hidden layer is varied from 1 to 70, and configurations with 1, 2, and 3 hidden layers are tested.

To evaluate each configuration, 5-fold cross-validation was performed on the training set, providing an average accuracy for each parameter combination. Cross-validation was chosen because it offers a robust evaluation of model performance, reducing the risk of overfitting by testing models on multiple subsets of the data. The optimal parameters were selected based on the highest cross-validation accuracy.

*2) **Baseline model Comparison**:* This experiment evaluates each optimized model's performance. The FNA dataset is divided into 80% training and 20% testing sets. Performance is assessed using accuracy, precision, recall, and F1-score. These metrics are calculated as follows:

- **Accuracy** measures the proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP is the number of true positives, TN for true negatives, FP for false positives, and FN for false negatives cases.

- **Precision** calculates the accuracy of positive predictions:

$$Precision = \frac{TP}{TP + FP}$$

- **Recall (or Sensitivity)** evaluates the model's ability to identify all positive cases:

$$Recall = \frac{TP}{TP + FN}$$

- **F1-Score** provides a balanced measure by combining precision and recall:

$$F1Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

III. Results

A. Model Optimization and Generalization Testing

*1) **KNN**:* The optimization of KNN focused on determining the optimal value for *k*, ranging from 1 to 50. The results showed that cross-validation accuracy exceeded 95% for *k* values between 3 and 7, with the highest accuracy of 96.04% achieved at k=5 (Figure 2). Based on these findings, k=5 was selected as the optimal value, balancing high cross-validation accuracy with stable performance.

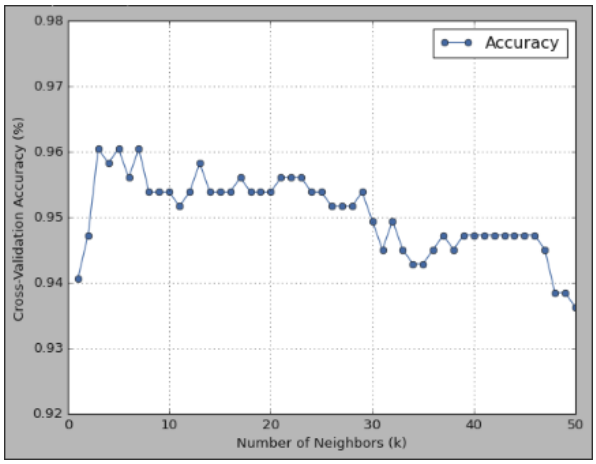


Fig. 2: Cross-Validation accuracy as a function of number of neighbors (k) for the KNN model

*2) **RF**:* The RF model's accuracy is based on both the model's depth and the number of decision trees in the ensemble.

The optimization results for the maximum depth of the RF model are summarized in Table II. For the maximum depth parameter, the model's accuracy showed significant improvement from 91.65% depth of 1 to 95.82% at a depth of 6. Beyond a depth of 6, the accuracy plateaued, with no changes observed for depths ranging from 6 to 26, all maintaining an accuracy of 95.82%. Thus, the optimal maximum depth for our model will be 6, balancing accuracy and computational load.

Maximum Depth	Accuracy (%)
1	91.6
6	95.8
11	95.8
16	95.8
21	95.8
26	95.8

TABLE II: Cross-validation accuracy table of the different depths for the RF model.

For the number of decision trees (n_estimators), accuracy increased from 92.31% with 1 tree to a peak of 96.26% at 211 trees. It remained stable at 96.04% between 151 and 201 trees, with no significant gains beyond 211. Thus, the optimal number of trees was set at 211 (Figure 3).

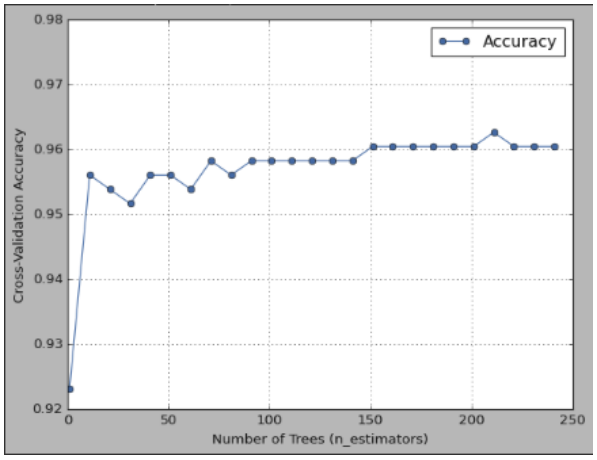


Fig. 3: Cross-validation accuracy as a function of the number of trees for the RF model.

*3) **NN**:* The optimization of the number of neurons per hidden layer was conducted by testing values ranging from 6 to 66 in increments of 5. While the highest cross-validation accuracy (98.02%) was reached at 46 neurons, there was only marginal improvement beyond 11 neurons (97.8%) with accuracy plateauing up to 66 neurons (Table III). Based on these observations, 11 hidden neurons were selected as the optimal configuration, as it maintained a strong accuracy of 97.80% while offering a less computationally expensive model.

The number of hidden layers was varied between 1, 2, and 3, with neurons ranging from 10 to 50 per layer. Results showed that a single hidden layer consistently outperformed configurations with two or three layers, likely due to overfitting in deeper networks and reduced generalization capability (Figure 4). Thus, our optimized NN model will use 11 hidden neurons and 1 hidden layer.

Hidden Neurons	Accuracy (%)
1	62.86
6	97.58
11	97.80
16	97.80
21	97.58
26	97.58
31	97.14
36	97.80
41	97.58
46	98.02
51	97.58
56	98.02
61	97.14
66	98.02

TABLE III: Number of hidden neurons vs accuracy on the NN model

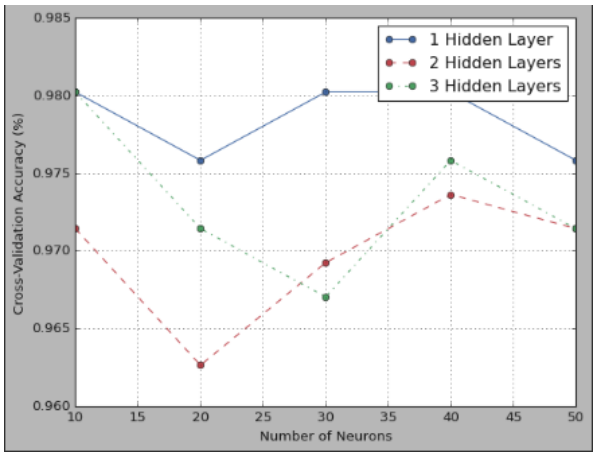


Fig. 4: Cross-validation accuracy for Neural Networks with 1, 2, and 3 hidden layers.

B. Model Comparison

To evaluate the real-world applicability of our models, we used test accuracy instead of cross-validation accuracy in this phase, directly evaluating performance on unseen data and real-world conditions. In addition to accuracy, we are also reporting precision, recall, and F1-score.

Using an optimal k value of 5, the KNN model achieved a training accuracy of 97.80% and a test accuracy of 95.61%, indicating strong generalization with minimal overfitting. The confusion matrix showed 67 true negatives, 42 true positives, and 5 false negatives, with no false positives. Precision was 93% for benign cases and 100% for malignant cases, while recall reached 100% and 89%, respectively. F1-Scores were 9% for benign and 94% for malignant, with an overall weighted F1-Score of 96%.

The Random Forest model, optimized with a depth of 6 and 211 trees, achieved a training accuracy of 99.56% and

a test accuracy of 96.49%. The confusion matrix showed 70 true negatives, 40 true positives, 1 false positive, and 3 false negatives. Precision was 96% for benign and 98% for malignant cases, while recall was 99% and 93%, respectively. F1-Scores were 97% for benign and 95% for malignant cases.

The Neural Network model, configured with 1 hidden layer and 11 neurons, achieved a training accuracy of 99.34% and a test accuracy of 98.25%. The confusion matrix revealed 71 true negatives, 41 true positives, 2 false negatives, and no false positives. Precision was 97% for benign and 100% for malignant cases, while recall was 100% and 95%, respectively. F1-Scores were 99% for benign and 98% for malignant cases, with a weighted average of 98%, demonstrating the NN model's exceptional performance in classifying FNA data.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
KNN	95.61	96	95	95
RF	96.49	97	96	96
NN	98.25	99	98	98

TABLE IV: Performance metrics for optimized models

IV. Discussion

The results across all models—KNN, Random Forest, and Neural Network—consistently demonstrate that the FNA dataset is characterized by simple, distinct patterns that do not require large or highly complex parameter values for effective classification. For KNN, accuracy peaked at k=5 and declined beyond k=20, indicating that smaller k-values are more effective for datasets like FNA with clear local patterns. Similarly, the Random Forest model achieved optimal performance with a maximum depth of 6 and 211 trees, beyond which no meaningful improvement was observed, highlighting the dataset's simplicity and the diminishing returns of additional computational load. The NN model further reinforced this trend, as accuracy plateaued with just one hidden layer and 11 neurons, demonstrating that larger architectures provided no additional benefit.

These findings suggest that the FNA dataset's simplicity allows models to perform well with minimal parameter tuning, reducing computational costs and avoiding overfitting. The consistent plateauing across all models highlights that simpler configurations are sufficient for datasets with clear, easily separable patterns like FNA.

Among the models, NN achieved the highest test accuracy (98.25%), followed by RF (96.49%) and KNN (95.61%). The NN model's superior precision, recall, and F1-score further underline its classification strength. NN's perfect precision for malignant cases (class 1) demonstrates its reliability in avoiding false positives cases, reducing unnecessary treatments. Its recall of 95% indicates effectiveness in identifying true malignant cases, minimizing missed cancer diagnoses. The F1-score of 98% highlights its overall classification strength, making NN the most effective for this task. However, NN's higher computational complexity and training time make it more suitable for datasets with intricate patterns or large feature spaces.

For simpler datasets like FNA, RF provides a practical trade-off between accuracy and efficiency. Its balanced precision and recall, combined with ensemble learning's robustness, make it more applicable in real medical settings. Ensemble methods like RF are effective for tabular data, effectively reducing variance and capturing diverse patterns through tree aggregation [9]. KNN, while simpler, serves as a reliable baseline when resources are limited and rapid deployment is needed.

This study highlights the potential for integrating machine-learning models into clinical settings to improve breast cancer diagnosis, reducing reliance on manual interpretation. Our models report higher true positive rates when compared to traditional, manual, diagnostic methods [10]. In resource-limited settings, they provide reliable and cost-effective diagnostic support where skilled professionals are scarce, offering second opinions or prioritizing high-risk cases. For example, an RF model could provide a practical solution for immediate deployment, while an NN model could be used for cases requiring higher precision with complex datasets.

While this study offers promising results for the application of machine learning in breast cancer diagnosis, several limitations must be addressed. The small size dataset (569 samples) may limit generalizability and underrepresent diverse patient demographics. Additionally, only FNA-derived features were analyzed, excluding insights from other diagnostic modalities. Future work should explore larger and more diverse datasets to validate these findings.

Furthermore, the exclusion of models like Support Vector Machines (SVM) and gradient boosting methods limits the scope of the analysis. Gradient boosting, for example, could enhance performance by sequentially improving weak learners, offering another dimension to ensemble techniques [9]. Additionally, investigating convolutional neural networks (CNNs) could also reveal how they capture complex patterns while potentially reducing computational costs compared to traditional NNs. Future research should consider these models to further optimize machine learning in FNA and broader medical contexts

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